

School Routine Generation Rules

Constraints

Rule 1 - Teacher Clash

A teacher cannot be assigned to more than one class section during the same period on the same day.

Invalid Example

Monday Period 2

Teacher: Anita Sharma

Class I-A -> English

Class II-B -> English

Rule 2 - Class Clash

A class section cannot have more than one subject assigned during the same period.

Invalid Example

Monday Period 3

Class VI-A

English

Mathematics

Rule 3 - Teacher Subject Locking

Once a teacher is assigned to a subject for a class section, all periods of that subject for that class section must be taught by the same teacher.

Example

Class VII-A

Subject: Mathematics

Assigned Teacher: Vikas Kumar

All Mathematics periods of VII-A throughout the week must be taught by Vikas Kumar.

Rule 4 - Teacher Eligibility

A teacher can only teach subjects and classes explicitly defined in the teacher configuration JSON.

Example

If a teacher can teach:

```
{  
  "subject_name": "Science",  
  "classes": ["VI", "VII", "VIII"]  
}
```

Then the teacher cannot be assigned Science for IX or X.

Rule 5 - Teacher Availability

A teacher can only be assigned during periods defined in their available_periods.

Example

```
{  
  "available_periods": [2, 3, 4, 5]  
}
```

The teacher cannot be assigned during Period 1 or Period 6.

Rule 6 - Teacher Daily Load

A teacher's daily allocation cannot exceed max_periods_per_day.

Example

```
{  
  "max_periods_per_day": 5  
}
```

The teacher can be assigned a maximum of 5 periods on any day.

Rule 7 - Weekly Subject Requirement

Every subject must receive the configured periods_per_week.

Example

```
{  
  "subject_name": "English",  
  "periods_per_week": 5  
}
```

At least 5 English periods must be allocated during the week.

Rule 8 - Class Working Days

Timetable generation must respect the configured working days of each class.

Different schools may have different working day configurations.

Rule 9 - Daily Period Limit

The number of periods allocated for a day must not exceed the configured period count for that day.

Example

"MON": 5

Only 5 periods can be allocated on Monday.

Priority Rules (Used During Allocation)

Rule 10 - Higher Rank First

Timetable generation should start from higher-ranked classes and proceed towards lower-ranked classes.

Example

XII
XI
X
IX
VIII
...
NUR

This ensures senior classes receive preference.

Rule 11 - Teacher Seniority Preference

When multiple teachers can teach the same subject, prefer the teacher with the highest class rank associated with that subject.

Example

Teacher A:

English -> I, II

Highest Rank = 5

Teacher B:

English -> I, II, III
Highest Rank = 6

Teacher B gets preference.

Rule 12 - Least Flexible Teacher First

Teachers with fewer available periods should be allocated before teachers with broader availability.

This reduces the chance of future conflicts.

Optimization Rules

Rule 13 - Teacher Subject Consistency

Prefer the same teacher to teach the same subject across multiple sections of the same class whenever possible.

Preferred Example

VII-A Science -> Deepak Anand
VII-B Science -> Deepak Anand

This is a preference rule and not a hard constraint.

Rule 14 - Avoid Empty Slots

The generator should attempt to fill all available class periods.

Free periods should only be created when no valid allocation exists.

Validation Rules

Rule 15 - Validate Before Final Output

Before returning the final timetable, validate:

Teacher Validation

- No teacher clash.
- Teacher availability respected.
- Teacher daily load respected.

Class Validation

- No class clash.
- Daily period limits respected.

Subject Validation

- Weekly subject minimum requirement satisfied.
- Teacher eligibility respected.

Final Validation

- Empty periods minimized.
 - All hard constraints satisfied.
 - Any unresolved conflicts reported.
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Class Rank Usage

The rank field in class configuration represents class hierarchy and weight.

Example

```
NUR   = 1
LKG   = 2
UKG   = 3
I     = 4
II    = 5
...
XI    = 14
XII   = 15
```

Higher rank indicates a higher class.

Rank is used for:

- Timetable generation priority.
- Teacher seniority determination.
- Conflict resolution.
- Teacher selection preference.